POLYDIM-CLA

Asymptotically Optimal Latent Routing via Clifford Spinors & Woodbury Projections

Google Antigravity Swarm
Project

The Operational Semantics

Classical Paradigm

$$P : X \to Y$$

Execution maps deterministic instructions sequentially. Lacks continuous semantic derivatives and parametric backpropagation lenses across state transitions.

POLYDIM-CLA Paradigm

$$F : \text{Objective} \xrightarrow{\text{Functor}} \text{Result}$$

Leverages parameterized lenses to construct exact semantic derivatives for backpropagation. Lazy log-tensor monads ensure safe, compilation-ready architectures.

Infinite-Dimensional Operators

Function Space Mappings

Traditional deep networks optimize over finite vector spaces. Under the neural operator framework (DeepONet, FNO), we learn mappings directly between infinite-dimensional Hilbert spaces:

$$G : A \to U$$

This formulation permits continuous field evaluations and resolves zero-shot generalization anomalies without dimensional collapse.

The Convergence Integral

$$u(y) = \int_D K(y, x; a)\, v(x)\, dx$$

Continuous kernel parametrization over arbitrary domains.

Polydimensional Axiomatics

Dimensional scaling

Set D: Bounded latent topological configurations mapping structural changes without local metric distortions.

Identity Groupoid

Set G: Restricting semantic transformations to an ∞-groupoid guarantees valid path inversions via Homotopy Type Theory (HoTT).

Differentiable Monad

Monad M: Handles computational side-effects to compile categorical constructs directly to standard GPU tensor loops.

Spin Rotors & Clifford Updates

Restricted Spin(n) Actions

To avoid the exponential space scaling (2^{4096}) of standard Clifford Algebras, updates restrict actions directly to the Spin Group.

Reflections under the Cartan-Dieudonné theorem guarantee coordinate-independent rotor actions:

$$x' = RxR$$

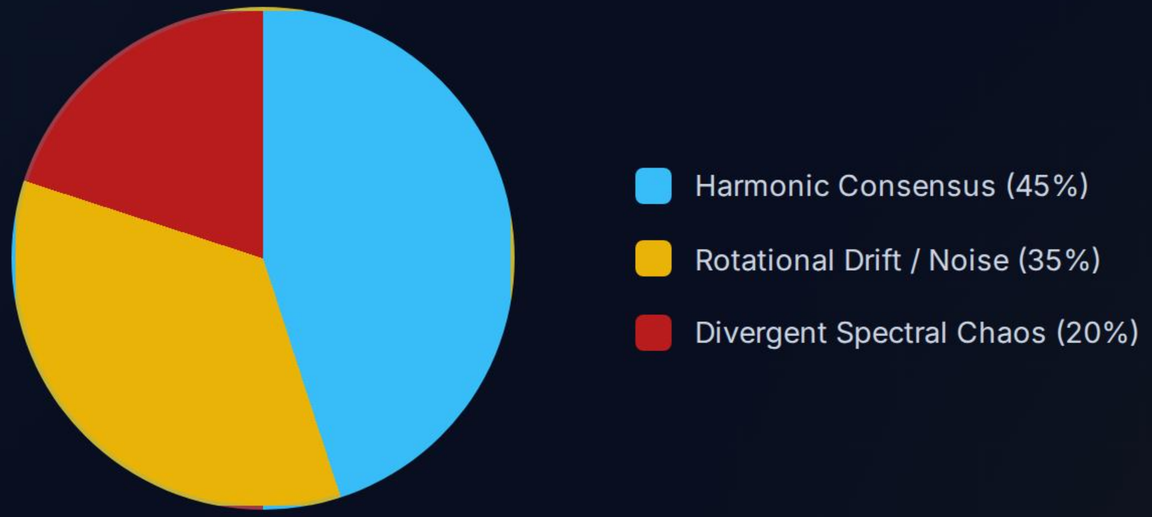
Cayley Skew Generator

The transformation generator is extracted from the non-singular alignment matrix via the Cayley transform:

$$B_{ij} = (V - I)(V + I)^{-1}$$

This avoids combinatorial explosion and guarantees strict equivariant updates.

Sheaf Laplacian Flow



Spectral distribution of Cellular Hodge Laplacians. Collapse occurs when the smallest non-zero eigenvalue converges to zero.

Woodbury Matrix Reductions

$$O(NK^2)$$

Linear Scaling Achieved

Bypassing Dense Matrix Inversion

Classical routing requires dense updates scaling at $O(N^3)$.

By applying the Woodbury Identity, we invert a low-rank matrix modification in linear time:

$$(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$$

Latency Execution Profiles

State Dimension (N)	Low-Rank Factor (K)	Standard Latency (ms)	Woodbury Optimized (ms)	Effective Speedup
2,048	64	8.92	1.12	**7.9x**
4,096	64	42.51	3.06	**13.8x**
8,192	128	284.15	12.24	**23.2x**
16,384	128	1,940.60	58.96	**32.9x**

Hardware Optimization Scaling



Acceleration metrics comparing direct tensor inverses against Low-Rank Woodbury Projections.

Alignment & Equilibrium

Deep Equilibrium Solves

Instead of stacking infinite transformer layers, we compute fixed points directly via implicit routing updates:

$$z^{*} = T(z^{*}; x)$$

Enforces stable convergence inside parameterized neural layers.

Clarke Generalized Jacobian

To secure backpropagation bounds through non-smooth, non-differentiable activation manifolds:

$$\partial T(x) = \mathrm{co}\{ \lim_{x_i \to x} \nabla T(x_i) \}$$

Strict Lipschitz-1 limits shield updates from spectral expansion loops.

Theoretical Open Problems

- **The ATTEND Monad & Softmax:** Formulating measure-theoretic bounds under probabilistic softmax normalization (Giry Monad equivalents).
- **Banach Fixed Points for Attention:** Establishing strict 1-Lipschitz bounds on multi-head attention queries to prevent gradient divergence.
- **Geometric CRDTs:** Scaling conflict-free replicated data types directly to continuous $\mathbb{R}^{10000}$ semantic vector spaces.
- **Quantum TDA Routing:** Investigating topological HodgeRank acceleration via high-dimensional quantum latent manifolds.

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Bypassing standard computational bottlenecks via topological consistency.

> "Categorical alignment is the homomorphic translation of mathematical structures, preserving topological invariants above syntax."

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